AI Bootcamp — Week 2: Software Development Workflow

Tuesday — Student Exercise Guide

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OBJECTIVES

By end of Week 2, the candidate can:

- Apply the EDD discipline to a domain with right/wrong answers: software development artifacts
- Build out the DLP student-workspace skeleton skills through EC-first iteration, adding Context and Constraints that are demonstrably load-bearing
- Explain the pipeline's YAML file contracts: what each skill reads and writes, the fail-fast behavior on missing inputs, and the review write-back loop
- Direct the AI to write a Python evaluation script against their EC (deterministic checks first, LLM judge second) and run baseline + iteration cycles with it
- Trace quality degradation through a multi-stage pipeline and isolate the source
- Produce an evaluation-measured iteration history showing net improvement across at least one full skill

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TIMING

- **Monday–Friday:** 9:00 AM – 5:00 PM with 1-hour lunch (7 hours each day)

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TUESDAY (9:00 AM – 5:00 PM): CREATE PRD: FULL EDD ITERATION

9:00 – 10:00 AM: HYPOTHESIS FORMATION

Review yesterday's baseline results for `create-prd`. For every failing check, form a specific hypothesis: *"This criterion fails because the skill does not specify X. I predict that adding X as a Constraint will cause this check to pass."*

Write the hypotheses before opening the skill file.

10:00 AM – 12:00 PM: CONTEXT FIRST, THEN CONSTRAINTS

Sequence matters. Start by adding Context to the skill: the WHY. The intent behind the PRD, who it serves, what decisions it exists to drive. Run the evaluation script. Record the delta.

Then add Constraints: the HOW. The data, rules, required sections, and never-dos the output must follow. Run the evaluation again.

This two-phase approach teaches candidates to distinguish "the model needs to understand the intent" from "the model needs more rules." These are different failure modes with different fixes. A useful diagnostic during the load-bearing audit: is this instruction in the right RTCC section? Intent language sitting in Constraints is a smell.

12:00 – 1:00 PM: LUNCH

1:00 – 3:30 PM: ITERATE TO GREEN

Continue EDD iteration on `create-prd`. Minimum: 4 total evaluation runs (baseline + 3 iterations) with a documented hypothesis for each change. One change per iteration.

After achieving Green (all checks passing): perform the load-bearing audit. Remove each instruction one at a time. If no check fails, the instruction is dead weight.

3:30 – 5:00 PM: WRITE EC AND BASELINE FOR ARCHITECTURE

Apply the same discipline to `create-architecture`: write EC by hand, direct the AI to extend or write the evaluation script, run the baseline.

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